Karl Friston's Unfalsifiable Free Energy Principle

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you know you're taking some of the 300 principle you try to think okay what's even more fundamental than that what's behind that what's what's below that what kind of feeds into that I have to say um also you won't get more fundamental simple in the free energy principle when you realize that your environment is composed of other things very much like you right when you're young your mum or as your older or your colleagues and your friends and your family and I think that's interesting because now we get into a much more symmetrical um sort of relationship between me and the environment and certainly in this context you know most of my universe is basically you and me so now there's a certain kind of symmetry in play um whereas you are my environment and I am your environment and yet we both have Geraldine models of our environments which basically means I have a generative model of you and you have a generative model of me the best way to minimize surprise when I surprising signals are generated by things like me mainly you is to ensure that I am as much like you as possible so that I can predict what you're going to do next because that's what I was going to do next and exactly symmetric is so for you so that basically means that we come to share a generative model a shared narrative we're seeing from the same film sheet and everything becomes mutually predictable what it doesn't describe though is the is the uh go back to your firelight Zone and the the man in the nice hell the epistemic hell so if it was the case that we could become completely mutually predictable I just keep on singing the same song together for hour after hour after hour that will become boring and at this point I think you're now into sort of social neuroscience and Joint free energy minimization um but you know can sometimes lead to quite paradoxical uh you know results or understandings of the way that we um search for information or engage with on the one hand I want to be make my world as predictable as possible so I don't get any nasty surprises so I'm going to go to those kinds of news channels I've talked to this kind of person because I think that you know we have a line frame of reference we have common grounds we speak the same language on the other hand I'm going to be compelled but slightly curious about other ways of thinking and it's not you know we're not all going to merge into one massive community and we are one massive hive mind one massive collective intelligence it's that's not going to work because we are compelled to you know what would have developed it this way or what's that kind of person like what's this culture like uh so yeah I haven't got the answers but there's a real you know really really interesting information in the in the in sort of social Neuroscience Professor Carl friston is a world-renowned neuroscientist and researcher he is best known for his work in developing groundbreaking statistical methods and mathematical models to understand the workings of the brain and how it gives rise to complex behaviors our discussion mainly tackles his free energy principle which states that the brain is constantly working to minimize the amount of free energy in its system according to this principle the brain is a predictive engine that is constantly trying to make script predictions about the world in order to minimize its own uncertainty if you got this far Please Subscribe and I hope you enjoy our conversation Professor friston thank you for coming on thank you for inviting me yes it is a pleasure um for the folks watching this uh on YouTube I just want to say they should know that you are one of the most sighted scientists in the world and have made you know Monumental impacts on the field of Neuroscience so it is the highest honor honestly to be speaking to be speaking with you today and thank you so much for coming on it's very gracious for you thank you yeah and um you know your time is very valuable so I do want to jump right into it and as we discussed over email the primary focus of the conversation will be on your free energy principle so I think it'd be awesome as as its Discoverer um let's maybe set the table so to speak could you provide a high level sort of simple description of the free energy principle and then I have a number of specifics to ask you about it right so when confronted with that challenge I normally say there are two ways you I know the description uh so there's a sort of um intuitive low road approach um which I'll describe first and then there's the appreciation of the principle from the through the lens of a physicist to try and understand self-organization but from the point of view of um a psychologist or a a neuroscientist we're talking about a very long-standing and a way of formulating our capacity to make sense of our world in terms of being able to predict so these ideas go all the way back to Plato and um probably kantian in nature but probably best articulated by helenholtz in in the 19th century that in the sense that perception is a constructive act it is um in the words of people like Richard Gregory it is the brain actively constructing explanations hypotheses fantasies for what might have caused my Sensations so this is a very much as Andy Clark would say an inside out process that you're creating a fantasy um an explanation for what I would have seen if this was the right explanation and then you're using the actual Sensations to correct update and revise your internal hypothesis your internal fantasies so on that view you could regard the brain as literally a fantastic all then it's a you're a purveyor of fantasies that are trying to make sense of of the lived world or at least the sensed World um and if you cast that in terms of inference mathematically um abductive reasoning and write down the rules that would have to obtain um if we were in the game of inferring the causes of our Sensations then you get to a formalism that can described in terms of optimizing a quantity called variational free energy and this quantity is quite ubiquitous and statistics and machine learning um also known as an Evidence lower bound and it simply scores the surprise that is inherent in any sensory evidence or any sensation and we use that surprise if you like to revise our beliefs and that mathematically can be written down as minimizing free energy so all that you can think of in terms of what constitutes a good brain and possibly a good mind can thereby be described as Trying to minimize free energy or minimizing surprise um you can understand that in terms of the average surprise which would be uncertainty so all that this is really saying it's a mathematical statement that self-organizing sentient systems look as if they are trying to minimize uncertainty trying to find the most um the best guess if you like the best hypothesis for the causes of their sensation you can understand this in a very generalized sense um you know you don't have to think of it in terms of sentence you could understand this is just a kind of homeostasis you know both your body and my body is in the same game of Trying to minimize surprising deviations from states of being and we do so um certainly on the level of physiology by trying to keep ourselves Within prior uh prior ranges of States a temperature or blood sugar and the like that would be very surprising if we found ourselves outside those regimes either very very cold or um and some kind of physical extremists so that would be if you like a sort of biological perspective on the free energy the physicist would come along and say um okay there are certain or possibly quite remarkable systems in my universe that seem to um restrict themselves to certain characteristic states of being um they have a peculiar capacity to resist the second law of Thermodynamics which I should say just applies to some closed systems but um you know I think it's used for a useful concept here in relation to this notion of homeostasis so what characteristics must these systems possess in order to keep themselves within characteristic States technically you know mathematically of course a pullback contractor these are just states of being that look as if they're attracting certain systems so that they gather themselves up and keep themselves Within These characteristic States and it turns out if you write down um the equations for the Dynamics of these kinds of systems then you can explain or even describe these in terms of a variational principle of least action that just says you're always trying to flow down gradients established by these surprises these prediction errors that's as Quantified by the free energy and that's a really nice oh that has a very nice interpretation because mathematically this surprise is the negative of what statisticians would call log evidence um it is a quantity that measures the goodness of your model or the negative of this goodness of your model which is just simply the probability of my Sensations given a model of how I think those Sensations were generated so this is exactly the same kind of generative model you'd find in generative AI or you know say large language models and the like so that that sort of gives you a theological gloss on this free energy minimizing process that's just a way of describing self-organizing systems that have this particular capacity to uh organize themselves and resist this entropic dissipation or dissolution Decay and death um and you can describe this as effectively looking as if they're trying to maximize the evidence for their models of the world so this um and I had to be read as basically Gathering evidence for my own existence if I am a model of my lived world also in Philosophy by um uh nicely denoted self-evidencing so it's a special kind of self-organization that can be read as self-evidising in the sense of gathering evidence from the way I think my world works and how I operate within that world so those would be two sort of uh perspectives on on the same phenomenon yeah it's wonderful thank you that's a great summation and the um we're gonna dive into this and many different we're gonna cut it we're gonna slice this up in my different ways but one thing I haven't heard and I've listened to many of your um you know podcast interviews and watch some of your presentations and read your papers but I haven't heard yet is what is the origin story of the free energy principle you know how did this all come together for you I know you have a long history um in the in the field but was there uh an aha moment or your Eureka moment some at some point when did this kind of all start to um come together what was the origin of this idea um well I think the first thing to say in response to that question is that this idea has been around for you know if not Millennials certainly centuries um and has been um increasingly formalized and and simplified so there could be many starting points to this story and each will have their precedence so this is a legacy um and a legacy of what well you could argue it's a legacy of um ideas proposed by Helen Hilton refined by psychologists Like Richard Gregory um Notions of analysis by synthesis and all sorts of um you know the brain as a constructive organ statistical organ trying to make a sense of its world that you'll find in the 20th century ideas which were taken up to great effect in machine learning by people like Jeffrey Hendon and Peter Diane they actually invented a helmhels machine you know that had this sort of um this Bayesian mechanics under the hood borrowing from another other Legacy left by Richard Feynman which was you know the the technical resolving the technical differences of doing this kind of uh this kind of inference hence the variation of free energy that he brought to the table um in the context of quantum electrodynamics um personally um it all started to make a lot more sense when reading the work of people like Raj Rayo um and Donna Ballard um and that story was a predicted coding story um but you know it transpired that this was just another way of telling the same story but you you are using a slightly different rhetoric so the prediction errors and predictive coding um just are a measure of free energy under certain simplifying assumptions but the key Insight um you know in the late 90s wasn't exactly the same objective function it could be used to describe both perception and the spirit of predictive processing as championed by people by Andy Clark and predictive coding but also the learning um the updates that you urge to connection strengths you know of the kind that you'd find in in machine learning but also you'd find in natural intelligence and experience dependent necessity in the brain and the um the thing that I found particularly intriguing was that if you just not stood back and I repeat so looked at this through the lens of a physicist you could also explain all of action through minimizing exactly the same quantity so um you know bit by bit everything that needed to be explained sort of fell into the camp of it's just minimizing free energy and everywhere you looked everything could suddenly be very simply explained in terms of minimizing free energy attention um uh and even in the past few years your natural selection is just another free energy minimizing process so this is um I haven't really answered your question in this house because there was no one moment sure every few years I I certainly get you know a nice aha oh it's just another instance of minimizing free energy and then you write a paper usually you know using numerical simulations and a bit of analysis to say well look this is another way of looking at it and it looks very much like this kind of thing because it's all um if you like a reflection of the same fundamental process of self-organization can you give us a sense of um I guess the scope of this discovery so as far as I understand it it explains um you know process a dynamic happening in the brain which is the most complex object that we know of in the universe uh what is the simplest scale what is the smallest um say dynamic or or thing that would adhere to this principle um does that make sense like basically what does this principle start yeah yeah no I just wanted to interrupt to congratulate you on the question because that's that's the second focus of many colleagues um in um very globology and physics so I'm thinking here people like Mike Levin and Chris fields and Jim glazebrook um so that is that that's absolutely key I think that notion um does this apply to a particular thing or does it apply to everything and if it applies to everything and what scale does it apply and does it apply at every scale and the answer should be it's scale invariant and it applies to everything literally um and that sort of um perspective is being championed um I repeat by my colleagues Mike German and and Chris in the context of quantum information theory on the one side on the point of view physics but on the other side um there's this story which I think is very nicely articulated by Chris fields that they're you know it is time to remove the bright lines between physics biology and psychology they're all just the same thing and you get this notion of basal cognition which is something I think Mike Levin has brought to the table that all uh self-organization can be read as this kind of basic sentience that's sort of basal cognition that is just an infection of of the existence of things and this may sound a little bit like an overstatement but um the you know the in terms of or the interpretation of self-organization and self-evancing is a relatively straightforward consequence of the individuation of anything from the rest of the Universe um and this separation um appeals to what is now a little literature or um on Markov blankets in this particular context so the Markov blanket is just a definition of thing that's where you start with the notion okay I want to explain the states of something and then you ask um well what's a thing you have to appeal to scrolling against formulation of you know the boundaries can you explain what goes on within the boundaries of living things and of course as soon as you say that you realize it's the boundary that's the important um structure that allows you to individuate something from everything else or no thing and indeed if you just look then look back with that um with that view on what you learned at school in terms of physics you suddenly realize how important that boundary is less than the heat path in physics for example uh so wherever you look the the there's this crucial notion of um a separation an individuation and simply by thinking carefully and articulating formally how you would Define that boundary um there is if you like a partial separation from states on the inside of something and states on the outside of something but there's a two-way traffic across this blanket or this this um you know this this boundary this Markov boundary and this two-way traffic enables you now to describe the internal Dynamics the Dynamics on the inside as holding beliefs about probabilistic subpersonal beliefs about the outside and it is that that enables you then to um assign this free energy functional to belief structures in the brain and you can do this for anything you can do this for a droplet of oil you can do it for a virus you can do it for a person you can do it uh in fact I've just signed off on a wonderful paper um by um my colleagues uh um and uh and Lance DeCosta where they've actually applied it to the biosphere so the same maths um unfolds once you've identified the separation of something for everything else you just apply this mechanics of spazing mechanics and and then you can interpret the self-organization in terms of sense making and self-evident and you know an implicit or a description of that sense making in terms of belief updating by minimizing the surprise or maximizing this basic Bayesian model evidence so um it can be applied to um single cells or it can be applied to the biosphere so it should apply to everything and so that's what Max Fields would say at what level of analysis when it comes to your Quantum information theoretic applications of the violation yeah it's fascinating the um I'm familiar with Chris Fields work I've read some of his research and actually spoke with Michael Levin a few weeks ago yeah we actually covered I mean in great detail um one of his papers I think it was 2019 paper called the the computational boundary of the self I don't know if you happen to see that but he has this uh this visual of cognitive light cones sort of the uh the boundary the assumed boundary of um of like nested cells let's say all going all the way down to the cell um say the cell then you have a second and then ant colony and it's sort of uh he brings up just fascinating ideas around collective intelligence all intelligence is being Collective and stuff like that so yeah a little bit that's those things quite a little bit familiar with um so it's great that you call them out uh can you define one thing it'd be great to get some definitions down can you define free energy so I believe in a thermodynamic sense it's the amount of available energy for a system to do work is a simple perhaps one of the simpler definitions for it is that applied in the same way in your free energy principle yeah again an excellent question um yes and no um so the you know the functional form of the variational free energy and that's why I tend to say the variation of free energy so that people do not misinterpret it as a thermodynamic free energy the functional forms are identical um and they inherit from the same kind of physics and probability um density Dynamics um however thermodynamic um uh free energy is I think a very particular application of this kind of mechanics to certain systems um so it is you know I think um from the point of view of a year so somebody is not a physicist I think it's important to say the variation of free energy is a an information theoretic measure it has no commitments to jewels or Boltzmann's constants or anything of that nature it's a it's very much like um a statistical measure like your a variance or a um an average or an entropy so the free energy as a free energy is just defined effectively in terms of a a relative entropy or sometimes known as a kale Divergence and a constraint or a um an internal energy um so the variation for energy is talking about um probability distributions it's talking about information and crucially it's talking about um information about things and then we come back to the the you know the central foundationals of the Markov blanket and separating the inside from the outside so the whole point of the three energy principle is that you can interpret the inside States as bowling information or probabilistic beliefs about outside States so this is if you like something quite fundamentally different from social Channel information on the surprise of finding the brain in this state so the brain will have two kinds of um free energy it'll have a variational free energy which is a comment about its goodness of fit or its beliefs about what's going on outside which a statistician would understand but it will also have a thermodynamic free energy in terms of um the actual metabolism and electrochemical signaling and and the metabolic states of of uh of the world that requires a further commitment to um I repeat sort of interpreting certain quantities um you know say the amplitude of random fluctuations in terms of thermal construct and applying things like Boltzmann's constant to it so there is a difference you know there are um between the two having said that I think that difference is specious on two counts first of all um you know taking a again a fireman s uh point of view information is energy and energy is information and at the level of the analysis that we're talking about um you know a thermodynamic interpretation is just another interpretation you don't need to understand self-organization um if you wanted to talk to some physicists in the 19th or 20th century will be useful but it's not necessary um actually another the other the other uh uh point is if you did talk to that kind of thermodynamic um person um then you would find that there is a um these three energies share the same minimum um they um there is a necessary thermodynamic cost to any variational free energy minimization or believer dating via things like that was principle so there is a physics which really binds these these two things together the final point of contact UM or perhaps the final um you know some of your more technical viewers um the free energy principle that we're talking about can be regarded as dual to James's maximum entropy principle so um James's maximum entry principle um basically said that um the universe or anything in the in the game of measuring anything else um will will conform to the principle of a maximum entropy under constraints and those constraints are what are supplied by the generative model that generates the predictions that are necessary to provide the surprise so the surprise part the internal energy part of the free energy principle is that which rests upon or is defined in terms of our world models our our gerited models so we come back to this notion the what is the fun what is the reality well it's an internal energy minus an entropy um so we want to minimize your energy you're going to maximize your entropy and Accord we change its maximum entry principle but under constraints that you're trying to minimize internal energy and that internal energy just is a surprise that we were talking about uh who we were talking about before um so on that view I think there's a your deep connection between uh a certain kind of free energy in physics but it would have to be a genesian the free energy even with the kind of strategy he did at school when he boiling water at night sure yes okay it's so interesting and actually the the you bring up William James's Maxim entry entropy principle something that I learned about listening to one of your lectures because I believe you said I think it's a quote if I have no preferences at all then my best bet is to keep options open so it's a bit of a size as that's like the non-technical kind of Layman's way to look at it um which I hadn't thought of before and I was actually thinking and it's one of the questions that I that I have for you and it's something that I think is really interesting about this principle is that um please correct me if I'm wrong or you know fill in here the preferences are are built into the Bayesian beliefs is that correct yes you you well you you've immediately gone into um a much more glorious world of preferences and purpose and plans and intentions yeah you know sorry we can go there later if you want because I it's no no no no it's a great place to be but just uh yeah at the moment we've just been talking about self-organization surpriseation self-evidencing but and what you what you've done is is really speak to um the active part of this um so you're just a set of scene um when you look at particular certain kinds of systems and specifically systems that have very precise internal Dynamics so the random fluctuations are suppressed or at least at the scale of observation that you might want to apply either all the random fluctuations are ambitioned away but these kinds of systems basically systems like you and me um but not um systems that are very very small and very very hot thermodynamically that would have lots of random fluctuations but if you can get a big enough system like you and me um then it is the case that the solution uh or um the variational principles that uh have to be in play in order for this um this separation to persist um the separation of self from non-self uh to be in pain um and you apply it to the way that you act upon the world then it looks then you can describe this in terms of minimizing um the free energy that you would expect if you pursued this course of action and it's at that point I think you get Notions of preferences and um all the how you could interpret this expected surprise and you know very much as we've said before um one way of reading expected surprise guess is you know in terms of entropy um but there's another way of avoiding surprises which is just to avoid surprising outcomes which means complying with the constraints that you know they were implicit in Genesis constrained maximum entry principle the complement of avoiding surprising outcomes is that it looks as if I am always acting in a way that um invinces preferred outcomes outcomes that just are the characteristic states that Define the kind of states that you'd find me in so I think you're absolutely right that you you know these prior preferences are just a way of articulating the typical or characteristic states that Define who I am so you can either sort of put a teleology on that and say oh this thing is behaving as if it prefers to be in these states it's actively choosing these courses of action these paths uh into the future that result that there's consequences are that you end up in your preferred States the physicist point of view you put it you you turn that on instead you start off with a system that seems to be able to occupy these preferred characteristic States and then you work out what Dynamics must these systems have so you know that's the deflationary part of the free energy principle you know it's just a description of things that have preferred States effectively and these preferred states are literally the attracting set that constitutes the pullback attractor that was mentioned before so you're absolutely right uh in one sense that these these preferences are part of Bayesian beliefs but that does if you like um commit you to a rather technological interpretation of the energy principle you don't have to do that you can do that you don't have to do and please forgive my ignorance yeah because I might uh I might be uh showing my hand a little bit here but is the brain trying to minimize free energy because it's trying to use as much of it as possible I know it's trying to minimize surprise and minimize uncertainty but like I imagine something like a resourceful greedy brain um so is is the minimization of free energy sort of like the byproduct is this a downstream effect or is it the the the the real pump that's that's moving the water along does that make sense it makes sense it's exactly the question which um I I try to intubate there there are two completely contradictory answers to so if you're talking to a physicist but no no no no no there's no sort of and there's no there's no aspirations to uh deploy my free energy in the most efficient way it's just that's what systems do systems just settle down to a free energy minimum that you know well let's put this another way those systems that settle down to an attractive set into a free energy minimum that that is their definitional attributes and stiffly that is what things are um so it is no surprise that it when um describe when you describe their Dynamics um they'll look as if they're trying to minimize their free energy because that's what defines them but if you now talk to a biologist um or a psychologist then your description is perfectly out yeah it will look as if these systems and particularly sort of uh um systems like you and me um are actively Trying to minimize a surprise and um you know in a resourceful way um I I would actually frame that more in terms of um in a very efficient way um so you know you know if you go to Wikipedia and you say well free energy is the amount of work available to do work I mean that that's a thermodynamic interpretation and you know I don't think it's quite necessary to to put that you know that's out of authorized things um but it certainly I think would be would be you'll be perfectly licensed just to interpret um various parts of or decompose variation of free energy and as for what would it what would it look like if we just focused on this particular part or that particular part and I think one really interesting decomposition of um variational free energy is that with a statistician would usually appeal to and that's basically um accuracy minus complexity where accuracy is effectively um how accurate your addictions of the world are how good is your explanation on these sensory data or if you're a you know a statistician sort of you know empirical data that they're trying to make sense of the other thing the complexity I think is is even more interesting um so in the same way you were talking about before keeping your options open and maximizing your entropy in a quarter of a change this maximum entry principle um the same sort of occams like um emerges in terms of this kind of complexity term so the complexity is just a way of lumping Together part of the internal energy with the entropy to give you this relative entropy or this Chao Divergence and to put this very simply what it measures is effectively the degree to which some sensory data or some Sensations or some sensory evidence changes your mind so if you're trying to describe this as a statistician this is the degree to which you move from your state of primarilies to your state of posterior beliefs so it's degrees of freedom you're using up in terms of providing an accurate account of your data and it is this that is minimized remember pre-energy equals accuracy minus complexity so I'll actually the other way around negative three energy is equal to uh accuracy minus complexity so as you're minimizing the free energy you're also you're trying to maximize the accuracy but at the same time you're minimizing the complexity so this is where I think you get what you what you were alluding to you're trying to accurately navigate the world in the most efficient way possible using up the least resources so why do I say resolve as well because of this um link between the information and energy you're afforded by things like the jinski equality and the endows principle it costs energy to change your mind it costs energy to create and well to destroy information so that complexity is also a cost technically it's also an information game and it's also um a complexity cost so what it looks as if a brain of this kind would be in the game of updating itself changing its biophysical state in a way to make predictions that are that as accurate as possible whilst at the same time complying with complexity constraints so doing them as parsimonious way uh that it can so hence Occam's principle um so that would also translate into um the most energetically efficient way of making sense of the world um and um you know if you sort of um that in support um in terms of computation in Silicon then what you know what that tells you is that the good computers um are those computers that work on the edge very cheaply with small amounts of electricity they're doing it very in the simplest way possible and be a with a cool head literally so I you know these these These are really interpretations of if you like unpacking the terms that go into a variation of free energy and making sense of them and I think it's a I think that's exactly what should be done it's a wonderful game um and you could also repeat that game in terms of the expected free energy so now you have expected accuracy and expected complexity and when you look at the functional forms of these terms you suddenly see things that people have been dealing with and optimizing or nearly a century so for example the expected accuracy um for the negative expected accuracy is just the ambiguity or the um the um the negative Precision um that would be used to explain an efficient sampling of the world of the kind that you might find in the say the visual cortex of the organization and the visual brain the expected complexity becomes risk of the kind you'd find in economics it is basically um the um the distance I move from my prior beliefs if I pursue this course of action and of course we've just said that the pride beliefs about the consequences of action are our preferences now describe our attracting set or our preferred States so now the expected complexity the risk is just scoring the degree to which I expect to be displaced from my third states of being probabilistically speaking let me just measure that with the cow Divergence on the relative entropy so that's you know that's um bread and butter for for many economists another way of um you feel like I'm packing or rearranging the terms in terms um is in terms of um the the what is referred to as expected Information Gain and expected cost so um those two um components are exactly those things you find in in the two aspects of being base optimal so there are two ways of being base optimal one way is to minimize your expected cost or maximize your expected utility or valuable log preferences and that's the kind of beta part you'd find in Bayesian decisions Theory and you can look at that as grab fathering things like reinforcement and learning and utility Theory um on the other hand though you've got this sort of again this sort of uh it's not ambiguity it's ambiguity with bells on which is the expected information game and that was exactly the objective function um devised at the Inception of the principles of Optum Bayesian design so if you have the problem of designing an experiment that's going to yield some data uh and you now ask yourself the question what's the best kind of experiment I could possibly design that will give me the kind of information that's going to resolve the most uncertainty about my hypotheses what's going to maximize the expected Information Gain and it is exactly that quantity that part that epistemic value in this measure the resolution of uncertainty that defines Bayes optimality in the context of experimental design that may seem a little bit sort of unconnected to you and me in conversational just walking down the road but of course we are experi we are actively experimenting all the time just by looking around you know the way that we move our eyes socatically every 250 milliseconds every little Act of this kind is an experiment so we come back now to the notion of Richard Gregory that you know perception just his hypothesis testing what's the experimentation it's just palpating the world in the right kind of of way to get that kind of information that's going to resolve the most expected surprise or the most uncertainty and you can have that you know at the level of visual palpation every you know four times a second or the news channels that you you you you know you would afford epistemic trust which is probably quite a hot issue at the moment or you know the kind of wicked you know whether you've got a Wikipedia or whether yeah what's what's who would you consult your your we make choices all the time in terms of what's the best way to act um and what are the and you know what the this notion of expected free energy brings to the table is that there are there's the imperative of a dual aspect it's pragmatic aspect that is um formerly this expected cost where the cost of the negative or the surprise in relation to my preferred states of being um but also to my mind even more important is this epistemic part this sort of expected Information Gain that actually makes us all little scientists of one sort of yeah that's incredible I'm glad I'm recording this conversation because I'm gonna have to listen to that a few times over to fully fold together all the pieces because there's so much there I actually found this when I was doing research on the free energy principle that I was going off and finding definitions for things because to bring it all together is um well it's amazing that it incorporates so many different um ideas from different disciplines I mean you have your Neuroscience but you also have information Theory I mean it's physics you know you have all these different things that you're pulling together um and finding these deep truths that I think are is fascinating um because that's exactly what Chris Fields wants he wants to sort of specifics psychology biology they're all the same thing and then again you know when I was trying to answer your question when was the aha moment that really wasn't one but every every few months you'd say oh that's just one of those well that so you know to my mind in a sense um it is that capacity to accommodate very gracefully things that definitely have worked in the past um that haven't been framed in this sort of simplifying uh sort of unifying framework it's that capacity which lends the free energy principle of a veracity and a utility that you know it continually impresses me that when when you were able to delay something that's endured for centuries or at least decades that we all use and work and that you know intensify understanding and suddenly it fits in with the free energy principle that to my mind is um and you know a reflection of the utility of the free energy principle but also interestingly um it means that the free energy principle is conforming to itself so if you remember the whole point of view engine is running accurate explanations as simple as possible so it should provide a simple explanation for everything that's that's that that's it's uh if you like um that's its foundational premise uh you know so you shouldn't really be surprised you can you can do economics and reinforcement learning yeah whatever yeah one of your my favorite papers is yours it's I think it's the 300 principle Made Simple made it simple simple as possible but no simpler I think is the title or it's close I'll link to that in the description for folks who want to uh to want to read that um I may have butchered that title sorry but it's close it was Einstein's famous quote yes it's from Einstein yeah and I even found that I didn't know this but there's a great little article in Wikipedia called uh the school the principle of simple Simplicity um so there's actually you know people like Nick chater and colleagues um have have found this you know this formalization lock comes principle and it's not you know I think the 300 principle could also be read as a duel to that kind of uh that kind of imperative to find those simple minimally complex explanation system yeah so I'd love to um I'll have to focus some more time on surprise the idea of surprise um it's Central to this Theory and correct me if I'm wrong so it's we're trying to minimize surprise or slash uncertainty right does the system want zero surprise precisely zero surprise or that might be I imagine that might be impossible actually so is it is it trying to approach zero is that optimally is that better does that make sense it does but again you're you're asking as a biologist not a physicist um sorry [Laughter] but there is a lovely Paradox here the notice we've been saying uh um in the past exchange that we are compelled uh if we exist as certain kinds of things that have these very precise internal um to maximize Information Gain so we are compelled to seek out Nova novelty we are compelled and seek out surprising informative um uh informative uh outcomes um which needs to be contrasted with what you've just brought to the table which is the you know the underlying the underlying imperative really is that of homeostasis uh to minimize surprise so you've got this interesting um dialectic here which just falls out of the maths that in the service of Trying to minimize my surprise to maintain my homeostatic to keep myself within these um Within These attracting sets that have a low entropy remember that the entropy is just the average surprise so if I spend my life minimizing surprise I am by definition when averaged over time minimizing my entropy is minimizing this you know the number of surprising physiological and mental and emotional states that that I experience during that period of time but to do that have to actually go and minimize uh maximize my expected information again so I'm going to look as if I'm actually very curious so part of keeping my home keeping myself out of unpreferred states and minimizing the surprise which you can read just prediction error for example part of underwriting my capacity to minimize my prediction error is actually to go and seek out surprising or apparently surprising um outcome so I I will I will be quintessentially curious I will be sensation seeking I will be answering to ask answer it'll look as a I'm going to want to answer the question what would happen if I did that um so we it is you know when you say we all want to minimize surprise we certainly want to avoid surprising Sensations yes um you know nociception the underwrites are um you know uh beliefs about being in pain for example uh being very cold being very poor being very unloved being dead you know these are all very surprising states of being which we will in the moment avoid but that does not mean to say that we will not seek out uh those outcomes that can have that have information so we're still very very curious and I I say that at length because there was an interesting uh philosophical Paradox called the dart room Paradox that was brought to the table around the Inception of the three energy principle in neural philosophy um and it went along the following well perhaps you know this you probably know the specimen I do do you want to tell tell your views on you also no I wish I did I'm sorry I don't know about it right I mean it's a very common sensical argument that you know if it's the case you want to minimize surprise why don't you just go into a dark room turn off the lights lie down stay there forever okay yeah um that's kind of an idea a question I had why not just jump off a cliff right that's you'll know the surprise will be over and then that'll yeah for extreme but yes well it's it's interesting that so so why don't we do that and yet we go bungee jumping I'm not jumping off cliffs and um bungee jumping uh you know they should be explainable under the free energy principle um and of course as a crucial there's a crucial difference between the bungee jump and the and the jumping over a cliff and look to come back to the dart room problem um I repeat it's just the Paradox that um why don't surprise minimizing machines or artifacts or systems basically sequester themselves from um their environment and close their eyes or turn the lights off and remove themselves from any particularly surprising Sensations and the answer to that well there are a number of different answers depending upon um how seriously you take the question uh well let's say this answer is well that's exactly what I do every day when I go to bed uh so it's not quite that surprising interesting yeah that's so good to counter yeah uh but uh but you know ignoring the fact that we do need some down time and there are good reasons for that in which come back to this um separation of free energy into accuracy and complexity um but more importantly the first thing I do when I go into a dark room imagine that you know Europe there's a conference and you go back to your hotel room and the lights don't work in your hotel room because you can't find the switch so you know what that tells you first of all the first thing you do you and I would do as free energy minimizing creatures in the future minimizing like expected for the energy maximizing my um expression information again it's a turn on the light yeah that's the first thing that any you know free energy minimizing system would do um which and the second thing that medical analogy tells you is that you are going to um you know if you can't find an idea you're just going to feel your way around so again this is expiration this is curiosity this is um the coming to the imperative to resolve uncertainty um remembering that uncertainty is average surprise so in resolving uncertainty you are minimizing your expected surprise but this is about um outcomes in the future that are not yet witnessed so again I think there's a you know when you um talk about um things I do then you are now talking about the imperatives that underlie action upon the world and the consequences that live in the future and in those instances it's all about minimizing uncertainty when you're talking about things um you know my states of being surprised it's just a measure of my state of being it's just the likelihood of these Devastation given my model uh my predictions of the sensations at this point of time point in time so the fundamentally different ways of using the notion of surprise I hope that makes sense it's probably a little bit over answered I've overshared haven't I no no it's great that that was useful I mean it is something that I think that's why I wanted to um remain at this on this topic on this on this level because it is I think the area where I do it I do still have the hardest time intuiting the difference between um Information Gain and minimizing surprise like those two things they seem counter or they seem like you know they CA you can't have both right um but I think it's partly perhaps a problem of language and in terms of like assigning a positive or negative negative valence to the term surprise so something like um like an example so I was thinking about this um the other day if if you won the lottery if you had a winning lottery ticket right you won a hundred million dollars um that's extremely surprising and also very good right but it seemed to be counter to minimizing surprise right um so but I think it's a difference of scale perhaps like um it's like the it's the difference between perhaps the overall model and a particular event within that model you know you know it perhaps using that example could you possibly help um help me into it sort of the difference and where where these tensions kind of resolve um but I think I think you've identified it well actually it's a wonderful let's talk about winning the lottery which is very surprising um yes I love I love to someday but also I think just uh you know just to reiterate what what you've implicitly just said that yeah the word surprise is applied to many different um constructs uh sometimes more anthropomorphic and sometimes less so but um when the surprise is used in the context of uh the free energy principle or um and mathematically it has a very specific and very deflationary meaning it's just the negative log probability of some data given your model of how those data were caused so it's a measure of the implausibility of getting these given a particular commitment to how you think these data were generated so I repeated probably best thought of as a predictionary you know um yeah I've some outcome um and I had a prediction on that outcome and the surprises is the measure of the mismatch or the distinction discrepancy between my prediction and what I actually observed um but this is also known as um the pry vowel Al an information Theory uh a term coined by tribus um more formally more generically it's also called self-information it's just a negative log probability of an event conditioned upon the model or the context in which that event occurred um so I I think that's if if I was talking to you know students of physics information Theory I wouldn't use the word surprise I'd use self-information because immediately they know that they expect self-information is entropy of a Shannon of a shell and sword so when I talk about minimizing self-information I'm using that as a shorthand that's saying I'm trying to describe systems that resist a tendency to um to um that's dispersion that states so notice that this is uh again as a very very confusing again this is um if you like almost the opposite of James's maximum entropy principle and there's a reason for that because what we're talking about here are is the entropy the average self-information or the average surprise of um actual outcomes of real um deterministic variables that that constitute um technically the sensory states of my Markov blanket the the way that the the world impresses itself upon me so this has nothing to do with beliefs or probability distributions it will surprise afforded this outcome and I want to you know being a good homeostat to use as a real cybernetics notion I'm going to want to make sure that everything everything that the world impresses upon me is within the bounds or you know that something like me can take and this is just a statement basis again um so that kind of entropy we're trying to minimize so remember before we're talking about James's maximum entry principle but the entropy was about the beliefs about the causes it wasn't about the actual outcomes so this is again if you like um you know a dialectic or a paradox you know on the one Handler Trying to minimize the surprise all the self-information of my of uh the actual uh outcomes that the the world impresses upon me but when it comes to um the um the Occam's principle view of free energy minimization I'm now going to try to maximize the entropy of my beliefs about the causes of that so it is complicated uh you know and perhaps like one should apologize for that it it's the same kind of like dialectic that you get into if you you know if you commit to James's maximum entropy principles the right way of measuring things and understanding things then you have to say well hang on a second I thought the whole point of biological self-organization was to resist the natural tendency to an increase in uh entropy I thought the whole point of Schrodinger's formulation of life was to minimize entropy so who's right Schrodinger or James well they're both right it's just that the entropy of the things I talk about are completely different oh sure yeah yeah James and and and the three energy principle are um the entropy of measurements and beliefs Schrodinger and homeostasis um and uh synergetics for example um uh are are talking about the entropy of outcomes of of measurements of sorry of the data upon that you use to to um to make your inferences and build your beliefs and update your beliefs so that you know it's a fascinating sort of Yin Yang and it really forces you to think about the nature of these mathematical descriptions and to what they pertain so you know things like entropy and surprisal and self-information are just attributes of probability distributions so you have to say of what is it the outcomes or is it your beliefs about things and with that sort of in mind you know winning the lottery is that surprising well if you imagine it the probability of it happening to you being surprising I can see that you might think oh that's highly implausible so it's not um it's not uh it's it it is very surprising on the other hand if you believe that you are the kind of Lucky child that good things happen to him and which you have to believe in order to actually um self-evidence your way and particular way into being a happy Chap and the good things happen to you winning the lottery is not that surprising that's the kind of thing that happens to people like me I'm a successful New Yorker and you know I wouldn't be that surprised because that's consistent it's ego syntonic it's yeah winning the lottery is not going to put me in a a truly surprising state that if I had a car accident or I had a stroke or um you know my partner left me you know these are truly surprising things whatever Lottery is not surprising because it doesn't take me out of my comfort zone my attracting set or my pullback attractor furthermore uh you know well I'm not sure that this example works so well in practice but um certainly um you can now if you um win the lottery if you did actually win the lottery that um actually reduces a lot of surprise uh in terms of what you're going to do next um so that puts in a certain position with a certain latitude of ways forward and ways and ways of Behaving which actually resolves uncertainty about um you know whether you're going to be able to pay these bills whether you're going to be able to support an elderly relative whether you're going to be able to pay for your children to go to college all of these uncertainties are exactly what you're trying to actively resolve but if you're very rich a lot of those that searches resolve themselves so that kind of if you like expected surprise which is all about the surprise that in terms of things I'm going to do is actually resolve my winning lottery so winning lottery is both good at that simple level of analysis I'm sure if you actually speak to people actually won the lottery you've probably been a very different different impression but at least at least at this level of analysis it's both good and surprise minimizing yeah right it depends on the frame almost the frame of reference like which what state are you looking at it at are you familiar um this came to mind the other day as I was thinking about this principle um do you ever watch the Twilight Zone yes from back in the day there's an episode called a nice place to visit and spoiler alert for anyone watching but you know it's an old show so if you haven't seen it by now but basically in the episode a thief a robber dies and goes to heaven and um they're a gambler they like to gamble so in a casino uh basically the way it works is they win every game at the casino they're playing craps or playing Blackjack and every time they win right and the robber is happy you know he's like oh my God this is amazing everything I play is it's so great and then please stop me if you if you've seen this episode already but for folks watching um he's happy at first but then it goes a month later and he's sitting at the table and he's bored out of his mind right he's winning every time and he's just like ah I won again like oh who why even play right and it made me think about this I mean maybe this is completely off but in the fact that there's no more surprise in the games that he's playing right so he's lost all of the will basically to go on and of course at the end you find out he's not in heaven he's in hell right he's got everything that he wanted but it wasn't it wasn't the right thing that he wanted sort of you know the lesson the story I guess but uh maybe think about this a little bit because it basically the surprise is zero and actually all the valence it's it's only a negative outcome really for for him you know he hasn't he hasn't realized that um the Folly of his actions you know there's no more hypothesis testing happening here every every time it's the same thing and what a dull life to live it made me think about this a little bit and please tell me if that's anywhere close to put this idea I know it's art I know it's different it's television but no I think that's spot on I I hadn't seen that episode that's a marvelous moral story I know you're absolutely right so that this is um you know not only do you see this um in in Psychology but when you start to simulate um you know little animals minimizing the fearity you see exactly this kind of behavior and as you say it's just the fact that the expected surprise um entails the motivation to do those things that resolve uncertainty you know and if you're denied the opportunity because there is no more uncertainty to resolve there's no more curiosity out there there's no more epistemic avoidance there's no more expected Information Gain that you expect because you know everything and it just keeps happening the same again and again and again and again that would be awful um you know um and you know when you actually simulate it um you you actually can simulate little artifacts who just get so bored they do nothing um and essentially they have you know once they attain their preferred States they're maximized their expected utility or minimize their um expected cost because there's nothing to shape their behavior because there's no no reducible uncertainty out there there's no more um there's no more epistemic affordance there's nothing to do uh and you know they just sit there and do absolutely nothing and that must be an awful situation to be in uh and I say that because you know 99 of your life is not chasing um you know if you were a mouse cheese it's not putting yourself in a position where you're going to win the lottery it's actually indulging in your curiosity the very Factor you're doing these um these podcasts having this exchange is just an expression of the fact you are compelled to be a curious creature though this is part of our makeup and if we didn't do this then you know at least from a mathematical perspective it's highly unlikely that we would exist over um you know over you know an appreciable period of time so yes I hadn't I hadn't seen that episode but you know I love the twist that you end up in and end up with hell yeah yeah that's bad well he was a thief so he deserved it uh I do have so about the um let's see here I'm trying to figure out where we're gonna go the I believe there's two ways right to minimize prediction error um they were change the model to better fit sensory input and or change the world act on the world in order to better fit the prediction but I wasn't sure are these two mutually exclusive I mean do you have to choose between the two and if so how does the system decide which one to do right so um we're now at a sort of um more Elemental uh application the 300 principle just to understand um the fundamental the you know the fundamental drives to both perception and action and you've you face up beautifully but so you know if we now um simplify things and then just read surprise and free energy in the moment as prediction error then there are two ways as you say that I can minimize my prediction or I can either change my mind so my prediction error has become more like the sensory data at hand or I can act upon the world to solicit some more or a different set of sensory input that are closer to my predictions so what would that look like and what it would look as if um I am basically acting to fulfill my predictions which is what I meant before so if you weren't optimistic uh your successful New Yorker um you know if you didn't have that prior that optimism that sort of um um optimism bias that you know um then you you wouldn't be able to act in a base optional way because you need to fulfill your own predictions to survive the you know you know there's something quite fundamental about that at the level I think we're talking um the kind of self-fulfilling prophecies we're talking about are very sub-personal so all we're talking about basically is uh effectively reflexes so you know when we when we move when we talk or move our eyes what's happening is that we um we're sending predictions down to uh the ponti nuclear to the spinal cord about the kind of signals we're getting from the state of our body uh technically it's called proprioception and if I predict that I'm going to be moving in a particular way I would expect to get these signals from my muscles and if I don't get them there's a prediction error so by acting all I mean or all that one means in the setting is that you send these prediction errors back to the muscle so that it contracts to the right length so it sends the signals that you predicted so this is just like a thermostat it's just like feedback control that we find you know in in many many devices all you need to supply is a set point the temperature you want or the stress the length of the muscle you predict will happen and then the body does the rest so this is how a very sort of Elemental level of just moving around um you can understand Action Motor Action Motor being the use of particular kinds of muscles uh as minimizing prediction error or minimizing free energy so it's not quite what you know there's more if uh deliberative uh planned kind of action you know it's not how do I imagine myself behaving tomorrow or generally um unfolding this conversation we're talking about action in the moment uh as basically eliminating prediction errors reflexive very very very quickly so an answer to your question how do we decide which to do and are they um are they um mutually exclusive they're certainly not mutually exclusive in the sense that the whole point of um this explanation of action and perception and the action perception cycle is that they there is a a common theme it's just all driven by minimizing prediction error uh or minimizing free energy in a more job context however in practice I think you're absolutely right that we we do seem to be built to um switch between the two so in principle you could do them both at the same time in principle you could dynamically adjust your set points and you your uh your reflexes could try and fulfill those set points or those predictions uh and the two could go hand in hand and temporaneously but in fact in practice it looks as if the the there's actually a term taking um and it looks as if um certainly sort of larger mammals um have a very particular schedule of turn taking uh I mean this space sound very specific but I think it's really interesting that yeah and it speaks to the cognitive moment um so it looks as if if we just take say vision then um our vision it's only Active Vision um where sometimes described as active sensing um where we actually have to go and palpate and select little parts of the visual scene bearing in mind we can only actually see a very very small part of the visual scene without you know with high resolution enough for real representations so basically um although we think we can see everything we're not we've just seen little patches and we're knitting it together in our heads it's fantasy that I can see everything that's not true you could see something if you looked over there and that gives you the illusion that you can see everything in fact you can't uh so this is part of the sort of the bread is a fantastic organ having the fantasy it can see all around me uh you can't you have to go and get very the most epistemically Rich expected to maximize the expected Information Gain um parts of the visual field to knit together and to accumulate the right kind of um the right kind of information that builds your hypothesis about what the scene that I'm currently constructing in my head and this process happens very very quickly so remove our eyes about four times a second okay yeah interestingly well yeah it is amazing but it's also well I find even more amazing is that everything seems to be about four times a second and by everything I made if you're a mouse you wouldn't really be using your eyes so much you'd be using your whiskers as you go around your different Burrows or you're in the field and you whisk about four times a second if you are talking you produce little chunks of information called phonemes about four times per second when you're listening you're listening at about four times per second when you sniff you slip it about four times per second so in the way that we sample the way that we sort of um uh gather our information from the seems to have this very saltatory aspect that you're roughly up four Hertz yeah technically a Theta Rhythm we seem to be going to sampling the information but what even more interestingly during the action during the actual movement itself the way that you adjudicate between changing your mind and updating your your allowing your brain to um revise its beliefs by changing its neural activity to provide better predictions um during movement um you actually attenuate the consequences of that movement to enable you to act so this is a really interesting phenomena called sensory attenuation um but simply you know if I had the um the prior belief that I was going to lift my arm and I didn't ignore or attenuate the prediction errors that were coming from my arm telling me my arm is not moving then I wouldn't be able to lift my arm because those predictionaries would come and revise my beliefs no no my arm's not moving but if I can ignore the sensations that I am going to generate then I can use my reflexes to fulfill my predictions that my arm is actually raising so in order to move I have to attenuate the prediction errors that are if you like undermining my predictions and this is called sensory tenuation and it's beautifully exemplified in eye movements so yo if you if you now you know look at this finger and then it's the card to this finger whilst you were moving circading and moving your eyes from one finger to the other finger you were suppressing and attenuating all the prediction errors and I know that because you didn't see the massive optic flow the shift of the world induced by moving your eyes you can actually see it if you press your eye this is how the horse's uh um famous experiment if you just gently nudged the outer edge of your eye you'll see the world shift around huh I don't want to push it too hard I'm like like careful to the viewers at home don't poke your eye out uh but okay just jump around a little bit it does yeah so um that's because your eyes weren't causing the motion um and you were able to actually register and attend to that visual that visual motion that visual if you like retinal slip exactly the same signals were being generated when you were moving your eyes with your eye muscles and you didn't see that you didn't see the world jump right all you saw was my finger and then my finger again in a different position the world had not moved so you were actually attenuating your sensory prediction as your visual prediction errors during the movement uh that's called psychotic suppression and it's a remarkable capacity of the brain to temporarily suspend attention to self the consequences of self-generated Sensations and this happens everywhere It's a Wonderful explanation for um uh the phenomena when you know you've got two children in the back of the car and he hit me hard and I hit him and you know the escalation in the argument absolutely right because when they're hitting they attenuate the uh sensory information from the reports how hard they're hitting so they actually feel being hit as as more intense than they feel they're hitting so they just they're just like a little arms race let's ratchet up and this is due to sensory generation you could actually use that metaphor for a lot of political a political exchange is that is that related to I know if you've ever heard this I actually don't know if it's technically true I think it is I've tested it out I've asked people to test this out how you can't tickle yourself yep yep the same kind of exercise that's a great observation so in fact that example comes from people like Daniel walpert who I think is now actually in New York um and Sarah Jane Blakemore um and uh you know trying to understand the the remarkable um first of all why is it you know it is remarkable you can't particular yourself because you're attenuating the consequences of self-generated behavior so in a sense you know the fact you can't induce visual motion by moving your own eye with your eye muscles is an example of that but you can circumvent that by tickling your eye tickling your eye by using something you're not used to which is which is your finger so that you know that it is exactly that notion you can't tickle yourself which explains this sort of uh little little experiment it actually it divides by uh by hell health so you can perform on yourself the interesting thing is that um what Sarah and colleagues and Chris frith were pursuing um and uh super shergill um was that people with schizophrenia have great difficulty um in this kind of sensory attenuation in another Paradigm called the force matching Paradigm which um means that people um there may be certain um psychopathologies um that interfere with your ability to suspend attention to the consequences of your own actions so for example if you had in a speech uh and you heard yourself because you were not attenuating the consequences of your own speech you might then try to explain away these unattenuated auditory or pseudo auditory inputs as if somebody else was speaking and then you had it so this is the notion that Chris Ritter brought to the table as an explanation for auditory hallucinations if um you were able to sense your own movements um and were not you know if you moved and you were not able to attenuate um the sensations of movement um it might feel as if somebody had caused your movement so you've got this phenomena made acts so there are all sorts of very quite frightening experiences that would ensue if you failed to have this sort of turn taking between action and perception especially at this very very time scale um and you know some of these frightening consequences you know one could argue I've seen quite frequently in many dices in Psychiatry and indeed neurology for example Parkinson's disease um an abnormality of dopenergic neurotransformation where dopamine in this context is thought to affect the um the capacity to do this the you know this kind of sensory attenuation at a particular level in the motor system and therefore um subverting Your Capacity to realize your intentions to move so what would that look like what it would look like somebody who can't initiate an act um and that of course is one of the common symptoms of Parkinson's disease so you know this this adjudication between acting and perceiving sort of you know quickly going out there moving actively ignoring the consequences of that women until you get there and then fixating not moving getting the information sorting it out doing your processing and then starting again the cycle of action perception where you have actually got the separation in play is at least from a biological perspective you know um seems to be a sort of ubiquitous way that we actively sense our world or actually makes make sense of our world and can be and is very very Reliant upon this uh this notion of selective attention sensory attenuation to get that that dance between action and perception um exactly right sure that reminds me if I tell you just a personal anecdote from last summer it was a random day I was out in my fire escape eating smoke meal uh and you know I do this in some days and all of a sudden there was I live on a one-way Street but coming the opposite way was a police chase there was an s a dark SUV being chased by a cop car and I remember the first second or two where I was looking over at what was happening I didn't quite see the vehicles I saw what can I describe as like a fuzzy black cloud it took a second or two for my brain to catch up to what was happening and to construct a car being chased by another car it was very bizarre because I mean it was highly unexpected right it just doesn't happen every day and um I'm not sure if that's a common thing with people who see things that are extremely uncommon but um I did have that and perhaps as I was just getting into sort of the perceptual um science literature and reading about the stuff so perhaps it's hard it's hard to disentangle that from expectations let's say but I mean it really stuck with me as you know one confirming although it's well confirmed obviously in the science but that this is what our brain is doing is constructing you know correcting for errors because that's a you know highly uncertain or really surprising thing to happen and my brain in real time took a second or two to cohere and make that a make that a solid those solid objects yes uh that that's a wonderful story and and yeah yeah as you say I mean it really speaks to the you know the brain as a constructive organ um as an organ that that really has tried to actively make sense of what's going on and sometimes in um ambiguous situations that can take many many seconds so you you would also I think you you may be able to ReDiscover um that remarkable if you like suspension of of the fallacy that we can perceive immediately uh and sort of see see yourself seeing uh using things like ambiguous figures so it's like a physicists have a whole range of really beautiful experiments you know using ambiguous figures and different kinds of illusions that deliberately put you on the edge so that you're you're you you you have um experience the sense making and all the sort of settling down at one option or another option you know resolving the fuzzy Cloud into oh yes that's the percept that makes the most sense in this instance um and sometimes it's you know when you're looking at ambiguous figures for example sometimes it stops making sense and you cannot voluntarily stop yourself seeing the other interpretation so you're famous two faces are revolves uh um example so you know I think these kinds of experiments that kind of experience really do speak very powerfully to the kind of experiments you can do at home to convince yourself you've got a very fantastic organ but but it takes time to construct the fantasies in situations which you're not used to um either you'll wait until you experience a a card a police car chase or you can go and look at some uh sorry illusory stimuli devised by by a psychophysicist I I should just add though if my um as well colleague Alan Hobson uh was here he just remind you that you know if you want evidence that your brain constructs all your percepts um just think about dreaming yeah yeah that is such clear evidence that we you know our vision is actually something we build from the inside and it's just gently constrained by um in the form of prediction errors the actual Sensations uh then just think about your visual experiences um you know during um enjoying dreaming so I want to ask you about that the while you're dreaming let's say having a standard dream I don't know what a standard dream is but you know that's like having a typical dream where you're in it and uh you're not non-lucid let's say and you're out and you're on the beach or something is there aside and said while you're in the dream you're experiencing it you're not you're not realizing it's a dream you're not realizing that you're constructing the environment the beach so is there in a sense a Markov blanket your personal your first person experience in that dream and the dream or does that make any sense like am I getting fuzzy here I mean is there technically is it technically the same thing or is it different um I think I think technically it is it is the same thing um it's just you know you introduced a really challenging issue there that you know do you know that you are dreaming or not I mean sometimes you you do know you're feeling it certainly contribute to into Lucid faces also uh really working making up so I'm just thinking um you know a proper answer to your question would have to accommodate the fact that you um you perceive and you act in this world um and yet you do not you're not aware that you are um you are actually dreaming and that musting as you say induce a whole series of markup and because first of all you're sequestered from the sensorium so the you know the biological state of of dreaming rests upon a particular kind of sensory attenuation which is very enduring and lasts for many hours if you're sleeping properly but it's mediated by the same neurochemical the same modulator neurotransmitters that we were talking about before relation to Parkinson's disease different kinds but certainly you know basically the same kind of functional role so basically you're switching off sensory input from the periphery on the body um so you're now letting your brain Free Will in terms of generating predictions um and um resolving out of uh in a hierarchical context where you're resolving the self-generated predictions by other parts of the brain trying to explain them away so effectively you're rehearsing your prior beliefs about the narratives that you usually um you know or something constrained by the experiences you've had the previous day and yet as you say you're not aware that you are dreaming so there's certainly high levels of this hierarchical construct or uh say deep levels if we think of a centripetal like hierarchy or heteraki um that um that have dissolved that sense of presence and wealth word or no sorry that is associated with the reality monetary uh so yes not only is a remark of blanket established and that separated you from the sensorium but there's possibly another higher one that separated your normal um Reality Checking and self modeling from certain parts of the brain that are engaged in rehearsing particular fantasies that you know are experienced as dreams one of the things I was thinking about with your model but the principle is I understand what the system is doing um a bit in terms of like having its own internal States is the environment can we think of the environment as doing anything with surprise or are we staying agnostic on what the environment is doing right um so again a very leading question so I'm sorry good good I've just yeah there's a whole other sort of uh podcast and possibly I'm not the best person to have that conversation there are people who who speak I think in an informed and very fluent way about this perspective but you you what you are suggesting um is that one can appeal to the mathematical symmetry of this way of partitioning the inside and the outside of the Markov blanket which has a an exact symmetry so that technically uh certainly in the at the level of sort of basic sentence and sense making um of the kind we were discussing um earlier on um it is mathematically true that you can also you can also interpret the environment as inferring your internal States because you've got this generalized synchrony which is another manifestation of um self-informational surprise or surprise minimization when the surprise of the measure of your sensory um the sensory Impressions so on that on that view the environment um now is acted upon by you um in the way that from the environment's perspective render your actions sensory inputs and the way that the environment acts upon you is by providing you with physical Impressions but yes you know mathematically speaking the environment so it should certainly be modeling uh you um and you know there are various flavors um of that truism that you could pursue one would fully acknowledge that there is clearly a vast asymmetry between you and your environment the environments usually making bigger it doesn't always have the same delicate structures or the precise mechanics and would require the sort of planning that we're talking about in terms of the expected free energy so we're not saying the environment brands but in terms of still trying to minimize prediction errors and doing the right kind of learning about you I think there is good evidence well you can certainly read certain aspects of the environment as doing exactly that so the favorite example is the um the elephant path of Desire path the shortcut that is worn by repeated use across a grassy lawn um when you get into you know want to cut a corner to walking through the park or to your favorite coffee cafe um so one way of looking at this is this is something I've done uh or you know to my environment and I've done it with mic on specifics and I've created a little path to uh my favorite face or the race that is the favorite of me and Michael specifics from the environment's point of view it's learn a lot about the behavior of its inhabitants of its denisence so you know you think you are impressing yourself on the environment the viable points environments point of view it's just invincing kind of plasticity an experience dependent kind of learning where it's remembering things about you um and you can take that argument right through to um those kinds of parts of the environment that are actually constructed by con specifics um like traffic lights like signs that have a semiotics and so deontic Valiant so you could just you know you could say that the the lived environment that you would Express if you walked out onto the street it's all man-made and it's all built to um uh maximize expected Information Gain I mean what is a sign it is there to create an environment that states our curiosity in terms of making it easy uh to respond to epistemic affordances that you're underright are curiosity or um our plans to maximize expected information again but we build that so this then you get into the role of yourself into the game of cultural Niche construction and the notion of the designer environment from Andy park for example um so you know all of these you know I'm just summarizing conversations you could have probably with other people they want to the final kind of conversation you might have is that when you realize that your environment is composed of other things very much like you right when you're young your mum or uh as your friends and your family and your uh and your your your uh on specifics on the street um and I think that's interesting because now we get into a much more symmetrical context young most of my universe is basically you and me so now there's there's a certain kind of symmetry in play gelatin models of our environments which basically means I have a change of all of you and you have a generative model of me minimizing the surprise in the exchange is uh one description of um can be um understood as leading to a generalized synchrony literally are meeting their minds a um a mutual predictability the restful certain things such as communication and language and certain common frames of reference common common ground but you could actually argue that in this setting it is an inevitable consequence of joint free energy or surprise minimization um when we put two of the things that are sufficiently similar together um that this would be inevitable um that the best way to minimize when I surprisingly signal they're generated by things like me maybe you is to ensure that I am as much like you as you're going to do now it's because that's what I was going to do next and exactly symmetrically so for you so that we're seeing from the same him sheet and everything becomes mutually predictable and that would be one explanation for the kind of in-group formation and the kind of um you know the mechanisms you might find in evolutionary psychology and you know that could be described in terms of Niche instruction what it doesn't describe though is the is the uh going back to your

Twilight Zone and the the man in the nice hell the epistemic hell so if it was the case that we could become completely mutually predictable and just keep on singing the same song together for hour after hour after hour that will become boring so now we come back to um you know what would this kind of mechanics and this way of understanding self-organization look like when I'm packed in societies and in exchanges between different kinds of con specifics and you get into really interesting uh questions about polarization in group out group the sphere of ideas and spread of memes that can be simulated um you know and at this point I think um that you know can sometimes lead to people on the one hand I want to be make my world as predictable as possible so I compelled to be slightly curious about other ways of thinking um but it's not you know we're not all going to merge into one massive community and you know one massive hive mind one massive collective intelligence it's that's not gonna work because we are compelled to you know what would have developed it this way or what's that kind of person like what's this culture like uh so yeah I haven't got the answers but there's a real you know really really interesting information in the in the in sort of social Neuroscience sure it reminds me a bit um I think you've spoken to it about the Explorer exploit trade-off uh eetl a little bit because it's sort of this the balance between exploiting sort of the the nearest say most accessible forms of energy or say news news content but then also having this balance between doing that and exploring say other modes of thought or you know um exploring what else is out there and that balance that getting that that balance right and I think you said something to the effect once about the free energy principle that it gets was it solves it in the right order something to that effect um I think it's and the way I perceived it was or reviewed it was in order to know what's surprising how to know what's possible kind of does that make any sense is that yes yes absolutely so yeah I probably said it sort of dissolves the exploitation expression dilemma but you're absolutely right it does so in a particular way um and it does match on the in terms of the order in exactly the way you say so if I put um if I go to this gambler's heaven uh you know there's an enormous opportunity for me to resolve uncertainty and to indulge my curiosity then you know respond to epistemic affordances which is the expected information part of the expected surprise and the expected free energy and that's so I'm going to first of all explore all possibilities what is possible in this particular new environment I moved to a new city or I've got a new uh device in the kitchen so the first thing I'm going to do is explore by becoming a little scientist and trying out this and that and going there and seeing what happens until I have resolved sufficient uncertainty that there's no more you know reducible uncertainty there and at that point the component the expected

Information Gain will fall below the expected costs or the expected utility so I'll have innate preferences you know um that uh constantly you know put constraints on my expiration so um at the point where you uh you literally um your epistemic um expected Information Gain Falls below the expected um value then the expected value will take over and I will switch deterministically to exploitative behavior so that means the first thing I'm going to do is explore and then uh so that there is no um uncertainty left to resolve and then I will exploit so exactly secure the possibilities and then um and then just act in a way that is predominantly dictated by my prior preferences or my constraints another way of looking at that is we are primarily um surprise minimizing um or uncertainty minimizing creatures but we do so in the constraints the constraints are provided by our prior preferences or by our the constraints uh afforded by the fact that our being in this state would be very very surprising so you know one way of looking at this sort of resolution of the exploitation expert expression dilemma is that it's not a dilemma in any sense we are quintessentially exploration machines okay exploration curious artifacts but we have constraints that we can't become dead or disabled or dehydrated or dissipated uh we can't allow ourselves to to be damaged um but under those constraints our primary you know imperative is to explore unless of course there's nothing left to explore and then we get more yeah sure it's the trade-off are you familiar at all with um I believe it's it's called a number of different things but uh it's like a game theory question so they call like the marriage problem or the secretary problem you've ever heard this before yeah so um would you like to hear it I don't know but like Yeah so basically the way the game works is um if you're looking for you know partner a life partner and the question is how much how many partners how many people should you date before um settling on one person you know how do you know uh the available options I guess how do you know when to stop looking is kind of the question it's also called the secretary it could be called the hiring problem because sort of like the question of you know you you see one applicant you have to say yes or no to that one applicant and if you say no you you can't go back to them right um jilted X let's say you know you can't ever go back to apparently what's optimal is and you have to know it's what's interesting is you have to know kind of how long you're willing to date you know how many partners are you willing to sort of uh to suss out and the answer is you you you basically date and you say no to the first 37 of potential Partners or potential applicants let's say for a job and then after that point you say yes to the first person who's better than all the options you saw already it's kind of how it works right so it's like you have to date around you have to kind of explore but even if you find someone that's incredible by this model you you're not supposed to say yes to anybody until you've seen a little more than a third of the whole people that you're that you're trying to observe and that just makes you think about eeto a you know requires a certain amount of exploration obviously there's certain parameters here that are more firmly set than you'll find for most system environment Dynamics but that there is actually technically a mathematically precise sort of strategy that works across the board so I just thought I'd share that no I haven't heard that that's that's brilliant and yeah it reminds me sort of evolutionary Game Theory having sort of you know stay win so tit to tap like different strategies and yeah which Super things in an evolutionary sense yeah the free energy principle how it how it reacts to or how it's related to the Hamilton's principle of least action can you possibly make that help um make that link yeah well um so um the free um as we were discussing in Iran can be regarded either as a variation principle of least action or um a constrained maximum entry principle is depending upon um you know your your favorite way of understanding the math um so the uh the variational principle of least action just is an instance of how it's expensive Middle East action um technically what both when applying Hamilton's principal at least actions say to um mechanics or movement of massive objects for example what these principles prescribe as a um a prescription of the trajectory or path that something will take so you could apply um Hamilton's principle of least action to the trajectory of a ball if thrown at a particular angle with a particular velocity team from this point um and it would be the case that the trajectory coursed out by that ball would be a path of least action it would be the path that minimized a certain energy when averaged along that path so this is a path um interval between meaning sort of adding things up so the path of least action the path of least at least resistance if you like uh where the resistance scores the um the expenditure of energy in order to you know because it deviated From the Path um it is prescribed by the principle of least action and you can use that principle then to predict or to describe the most likely path uh that anything will take um and indeed that that sort of path Interpol view is exactly where free energy the variational free energy comes from but here we're talking about sort of Feynman trying to work out the probability of various Paths of small particles or say electrons um so you know a very difficult mathematical problem that was finessed by introducing this notion of a variational bound on the probability of a particular path uh in his pathological formulation Quantum electrodynamics the math is identical to the free energy principle and it just says that there is a particular path through a space of beliefs that is the most likely um and um in the same way that there is a particular path if I throw a ball that is conforms to help someone's principle release faction um the the the space through which we're um perhaps traveling now is the space of probabilistic beliefs encoded by the internal states of something so slightly more abstract it's more about the measurement thing um the um the sort of genes in use of the word entropy or the way that James would um apply his entropy to to belief structures um but the you know the the mathematics the variational calculus is exactly the same it just tells you given um a particular initial condition um uh and um some sensory input for example I can tell you the most likely trajectory of beliefs that would be described in terms of these Leaf updating for example um that it will will ensue using this variational principle of least action which is the free energy principle it's exactly the same man um yeah I'm just trying to think of even more intuitive ways of um of describing it yeah it's it's a path of least that effort it's just a mathematical way of describing the path of least um cut into the future or the past of of least effort where the effort is effectively effectively scored by the amount of surprise I can I I can um I can avoid gotcha and are you familiar having to be familiar with um Jeremy England's work uh with dissipative adaptation yeah yes well I was about four years ago pre-covered yes I haven't I haven't read what what he's been up to recently same movies oh really oh I didn't realize it moves yeah yeah but it's it's far more of it I know it's more of a thermodynamic it's solely I think I believe from thermodynamic but it makes me um because one of my uh general interest or one of the things I try to get to is like just you know you're taking something at the 300 even more fundamental than that right you can keep trying to cut away and say okay well how do you make this even you know what's what's what's what's what kind of feeds into that and um I don't I mean it to be completely transparent I don't fully understand how dissipative adaptation works I think has something to do with uh the environment uh adjusting itself to best accept the energy that's flowing through it some something like that but of course it's very technical um is that at least I know it's maybe been a few years since um you follow the uh follow the idea but does that does this conform well I guess it would have to but uh does this play into the free energy principle at all or is this a dynamic that kind of works it's fascinating work and you may you may also want to invite Suzanne still uh into one of these conversations who's been doing very similar work um from the point of view of sort of predictive information Theory and and recently she's also been looking at this sort of uh remarkable capacity of systems um to uh again resist the second law and in your your use use their free energy thermodynamically but she would frame it in terms of information Theory um to um if you like extract order from from chaos um and you know Common to a lot of these Notions some notion of a ratchet that you you know you you you you you you could make little random jumps that you you um you maintain them um and each jump uh is maintained um if it's heading in the right direction uh you know fail minimizing uh sort of uh free energy offense of something else however I have to say it's not really um addressing the same kinds of issues that the free energy principle is addressing I have to say or simple in the free energy principle um so I look at thermodynamics as one particular domain of application of that kind of mechanics um the only difference between thermodynamics and the free energy principles the free energy principle actually commits to a thinner so it commits to a Markov blanket so it's now got a partition of states where you can start to talk about beliefs in a non-trivial sense in the sense that um there is a um amorphism from the internal states of a physical system to the states beyond the heat path or beyond beyond the blanket and all the more interesting questions we have about maintenance of that blanket or the heat math it's not about the equilibrium physics of things on the inside most of feminism just rather uninteresting descriptions of ensembles of things on the inside that have some particular exchangeability criteria and then you can apply for classical equilibrium physics to that that's not terribly interesting from the point of view of um people looking at non-equilibrium physics and even less interesting from the point of view people looking at the real randomly dynamical coupled systems because these systems are not exchangeable you you mentioned before that Mike Levin's description of your cells of cells of cells of cells and boundaries about as a boundaries and boundaries Markov blankets and blankets and blankets and blankets there's a heterogeneity to implicit in that scaling variant description of actual systems that will not allow you I think to apply to adopt the assumptions that are that we would have to and commit to if you wanted to apply thermodynamics so I I am it's serve attitude to thermodynamics um in that sense um the you know trying to understand the thermodynamics of itself organization I think it's possibly misguided because you have to make too many commitments to the kind of systems particularly the exchangeability of ensembles to make it really interesting on the other hand the the questions that people like Jeremy England and Suzanne still more recently have been addressing and I think deeply interesting uh and I'm not really impressed by the free energy principle I think what you're talking about here is the there's a sort of uh the complexification that seems to be an inevitable property certainly of biotic self-organization You could argue other kinds of self-organization and certainly biotics of organization uh far from equilibrium systems that have this kind of scale freeness and implicit heterogeneity in them you know I think that there has yet to be um a simple story to explain why we are becoming increasingly more sophisticated increasingly but if we are we're organized uh uh I think that's a fascinating issue I I don't have a clear answer you know and I will be um the free energy principle itself does not properly address that because the Friendship principle starts by saying let's assume something as self-organized it's uh has an attracting set that's where it starts and that's really where it finishes um I think if you want to think about um you know how these attracting sets evolve and contextualize each other you know you're going to be in a slightly more um challenging frame which lots of people are addressing and it's a really interesting question I'm not sure thermodynamics is a right is the right toolkit to address that to be honest I think you need to look at more fundamental maths um probably important information so at different scales and think about that through the lens of say political biology and evolutionary thinking sure I'm so glad you read a Quantum information Theory because that's where I wanted to go next the the idea and I was wondering if the 300 [Music] lends any Credence to the idea or has any relationship to I'm not sure I'm not sure if you're familiar with um some theories about the universe being like a Quantum error correcting code uh if you're familiar at all with with that landscape um but basically it's you know it's a speculative idea but it looks like there's some evidence to support it that the the fabric of space time itself is actually like a Quantum air correcting code and when I hear that and I think about the free entry system Trying to minimize uncertainty it's trying to you close to air correction or it seems kind of similar too I wonder if there is a potential link there and if um if you have any comments about that or if you've if that's something that's that's come across your uh your desk one day it literally came across my desk about about um about three or four weeks ago so somebody sent me this paper on the University and I'm delighted uh and um if I remember the sentiment of the email was you know it would be great to get this chips or This Woman's um take on this you know in relation to a free energy theoretic formulation because you know it seemed as if there was a very very close connection uh uh as you as you have spotted I I think the best person to comment upon that would be Chris Fields um you know so um you know he you know I I did quantum physics as a young man that that was 40 years ago I usually defer to the dress when it comes to um Quantum information Theory so he's a he you know he he has a a beautifully rounded perspective on these things and we'll probably be able to give you quite a definite advance so obviously yes it's the same thing but they're trying to explain why it's the same thing yeah it's hard to like again I will be surprised if all of these things were weren't entirely consilient um yeah and I probably ultimately you know you probably aren't going to fall back on something like um James's maximum and support um certainly if you're in the public Quantum information Theory um you will probably find you know an equivalent um description of the kind we wanted to mention in terms of variational principles of least action but you may also find the similar um explanations in gauge Theory and you know and ultimately category Theory but they should all I think be internally consistent you should be able to sort of move from one to the other without doing too much established deal commitments in anyone your favorite way of framing them right I'm sure directing universe is is one example of this yeah it's very much could be it reminds me too of Christopher's a frequent collaborator of his is um the cognitive scientist Donald Hoffman I'm not sure if you're familiar with his work um I've interviewed him a couple times on this channel as well but uh one thing that you said that's that stood out to me and it's something that that he he said as well or something similar is that we don't need to know the truth we just need to minimize prediction error you know it's like this very separate thing I don't know if you um if you're familiar with he's got a few different theories I mean one of which um the fitness speech truth theorem that we've evolved in order to basically maximize Fitness instead of to see what's vertically actually there in the environment so much what we see is uh or all of what we see is an illusion we can never hit he posits that we can never actually get to the base reality or the bass truth and um I'm not sure if you have any thoughts about that or if it's something that um you've come across if you've spoken with Don no I think that's absolutely right yeah and it sort of leads you to interesting discussions uh should you want to go there in terms of skepticism versus realism to internalism versus externalism and the like with philosophers but yeah certainly from the point of view of the maths you know the physics of this kind of sentience it is all about uh making um forming or understanding internal Dynamics as um encoding some personal probabilistic Bayesian beliefs about stuff which is fundamentally hidden Beyond Your Mark of blanket beyond your sensory Veil it is inaccessible it is unknowable and I think probably Chris Fields would frame this in terms of holographic screens uh some of the people to the holographic principle that um you know ethical exchange of information across your sensory there or your Markov blanket can be construed as a holographic screen from interview of quantum information Theory and that's the only way you can entangle or couple two systems uh you know with classical information so you know the again we come back to this fundamental separation between the inside and the outside which you know read um in terms of your your conversations would be exactly you can never know what's out there that really don't need to you know if you can keep your prediction errors very very small for as long as you live job done I mean that's that's perfectly that's that's perfectly okay it does it does raise the interesting issue though is there an outside uh which which will be a more philosophical issue so both Chris and I I think would have to commit to the fact that it probably is an outside in order to uh have um to actually be able to read and write to your holographic screen or to be able to transact with an environment uh you know in terms of in both directions over your markup plan blanket so the strategy principle puts into slightly ambiguous position when it comes to philosophy so on the one hand you can be entirely skeptical everything that needs to be done uh is beneath your Markov blanket within the skull bound brain uh there is nothing else um on the other hand that you're you're making sense of signals that are provided by something out there so you have to you have to commit to at least a mathematical discouragement of stuff being out there whether something is sure I'm just looking at the time and I realized we've gone way over here we had sort of uh planned out to go I hope I hope you've enjoyed this discussion I mean I've gotten so much out of it um professor and thank you so much anything else you want to um I mean I I literally have so many questions I could continue to ask you but I do want to be mindful of your time and energy as well so please let me know if you know if you have a few extra minutes I can ask you a couple more but if you know if you've reached your limit please let me know also please ask ask me two more questions and then I gotta go and have a nice cigarette okay oh that sounds nice that sounds lovely um yeah one of those once in a while isn't so bad um I think okay okay two questions oh now I have to really narrow it down here we'll see it's a little bit um well one it was so one I'll ask you is and I'm surprised we actually haven't touched on it um active inference which honestly for I mean maybe the first couple weeks I was looking into the free energy principle I actually thought they were the same thing I actually think it's it's written in other places I think Wikipedia may be kind of writes them as equivalent and I believe active inference is a subset of the free energy principle or it's like um well perhaps I mean you could probably you could definitely do the best job can you help um separate the two or help us make a distinction between the 300 principle and what active inferences I think you could be forgiven by by using them synonymously and certainly you know that's how it was portrayed I think in in the in the Wikipedia links having said that um yo in the world of um philosophy um well it's got to be a little bit um more rigorous in the way that you um you frame various offerings uh so nowadays um and actually I am and a lot of my friends um in the philosophy are also very careful to distinguish between the free energy principle which is just a method so very much like we were talking about Hamilton's principle of least action as a method for determining the path of least action or least effort um the free energy principle is just what that what it is it's just a method it's a tool for describing the most likely um kind of active sense making or belief updating or or um or as you say active inference and so it's not falsifiable it's not a theory it's just a principle is this a tool it's a method if you apply that to um purposeful Behavior then you have active imprints so active infants I think is an application of the free um to um usually simulate or predict or emulate or to model um purposeful sentient Behavior basically so so I've given you a long-winded technical answer but so you you can you can so free energy principle is is usually for sort of physicists and philosophers that the fossils like to shout about it the physicists think it's probably quite quite trivial um the uh but active inference is more an application that you might want to actually apply if you're you're a a psychologist or artificial intelligence research and wanting to emulate this kind of behavior or understand say a given subject's Behavior by fitting an active inference model to that that subject's behavior using a much more magical way and you can regard as only having utility in that ability to simulate active infants just technically um the only thing you get from the free energy principle is really a way of writing down the Dynamics of self-organization as a gradient flow on the um on the generative model on those prior preferences on the proper description of those attracting sets um which means instead of just looking at the way a system behaves given its Dynamics and trying to reverse engineer its lagrangian uh what you could do is actually write down a little longer as a change in constraint as a set of Prior preferences and then just apply the rules of the free energy principle to simulate the behavior of a system that self-organizes to that set of attracting States on that particular set of preferred States and that's basically what we do in computational uh neuroscience and machine learning research okay thank you very much that helps the distinction is super helpful um right now last question uh did you I had to kind of bunch of different ones I'm like oh I forgot to ask them about attractor Landscapes and strange loopiness but I will instead ask you um if you this is more of a meta this is personal question but um if you could give your 20 year old 20 year old self one piece of advice what would it be so if you go back my 20 year old um I think I think it would um no I I I'm trying to be too clever about this probably probably to go to the dentist more often to me less trivial answer I I think it would be would have been nice to do a couple of years of philosophy as uh yeah I always say when people when my children and when my students and when other people ask me what you know what's good career advice is keeping your options open you use that phase earlier on today and I think there's something quite fundamental about that as broad found a foundation as possible bearing in mind that we are all quintessentially curious creatures and therefore if you commit to an academic career you are indulging that but the best way to explore is to have a you know a very broad Foundation um and I managed to do that to a certain extent but I I didn't I did miss out on a lot of history and a lot of philosophy and I remember one of my mentors Jerry Adelman uh used to refer to me and I won't say fondly he just used to refer to me as an intellectual Thug because of my lack of scholarly had very little finesse when it comes to sort of uh fantastic or history so it would have been nice I don't know what would have happened if I if I'd spend time doing that but it would have been nice just to indulge myself with a couple years in that kind of training sure to have that base but yes there's still time I mean it's not like you know I mean I'm sure you're busy so you don't have too much free time but you know it's always available foreign thank you so much this has been I mean this has been amazing speaking with you and um I'm so honored so blessed that you take the time to speak with me and I I'm sure the audience will appreciate um everything here and I'll include links in the description of this video for people who want to learn more who want to read your papers I'll link some of your papers and some of your um uh some of your presentations and uh yeah thanks again it's a treat oh thank you our village by talking to you and I'm going to find out how long we've been talking I deliberately haven't looked I shouldn't have said anything I wish I hadn't said anything now because I could have asked you about a tractor Landscapes but no yeah it has been a while and I appreciate so appreciate you going over all right I should go away thinking um attractive Landscapes and what I should have advised myself to do when I was 20 years of age that's a great question all right